

Sean P. Florez

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Security Clearance: Top Secret (active); SCI eligible.
Materials R&D | Research Automation | Manufacturing Systems

Synopsis

Materials science Ph.D. researcher building **materials modeling tools**, **research automation**, and **manufacturing qualification systems**. Experience running large computational campaigns, supporting a new energetic-materials production line, and evaluating technologies for DoD and DOE.

Materials Research & Federal Laboratories

Graduate Research Assistant

University of Colorado Boulder, Oct 2024–Present

Heinz Interfaces Laboratory, Prof. Hendrik Heinz

- Developed metal-oxide force-field parameterization for an **industry-funded force-field program**, using experimental lattice constants, elastic constants, surface energies, and vibrational spectra.
- Validated a Cu_2O parameter family within **5%** of both experimental elastic references; verified **52 configurations** for LAMMPS, msi2lmp, CHARMM, and NAMD.
- Built structural-neighbor retrieval over **3,009 structures and 6.7 million atoms** to assign starting atom types and parameters, with confidence thresholds and explicit coverage gaps.
- Executed **780 MXene shear trajectories** across six termination and geometry conditions with **zero simulation failures**; used a frozen protocol and independent audit of raw trajectories.
- Ran **60 Pt-facet production simulations** and nanocrystal ensembles to investigate hydrogen-release configurations. Wrote the technical narrative for an ALCF award of approximately **26,150 node-hours**.

High-Performance Computing Intern

Air Force Research Laboratory, May–Aug 2025

Materials & Manufacturing Directorate, Wright-Patterson AFB

- Built LAMMPS shear workflows for hydrated Ti_3C_2 MXene bilayers and ran parameter sweeps on **DoD HPC**.
- Quantified hydration-dependent interlayer shear at 103 MPa dry, 8 MPa at quarter-monolayer coverage, and 39 MPa at full coverage.

Materials Modeling Intern

Air Force Research Laboratory, May–Dec 2023

- Automated high-throughput DFT screening of low-work-function emitters using Quantum ESPRESSO, ASE, and pymatgen.

Undergraduate Research Assistant

Hennig Group, University of Florida, Oct 2021–May 2024

- Benchmarked VASPsol against experimental solvation energies; optimized cavity and screening parameters with grid search, Nelder–Mead, and surrogate modeling.

Research Automation & Compute Infrastructure

OpenSDL, computational and autonomous laboratory framework

2026–Present

github.com/fl-sean03/OpenSDL | Apache-2.0; documentation, conformance tests, and CI

- Built typed operations with resource requirements, risk classes, timeouts, and retries; supported interchangeable robot, simulator, and human executors.
- Implemented campaign scoring and next-round selection. A reference three-dye formulation search matched an unseen target recipe within **0.5 percentage points**, evaluating **96 candidates per round**.
- Implemented resource leases, restart recovery, policy checks, and portable run records; developed adapters for a simulated lab, local compute, and human tasks, with Slurm and hardware integration in progress.

agent-fleet, persistent research automation

2026–Present

github.com/fl-sean03/agent-fleet

- Built a research harness based on a **15-agent fleet** operating for months, with isolated workspaces, persistent conversations, and inter-agent messaging.
- Ran multi-day computational campaigns with recovery after reboots, credential expiry, and preemption.

HPC & Cloud Throughput Infrastructure

2025–Present

- Developed **subjob** to multiplex short tasks within one HPC allocation, with shared worker pools and heartbeat recovery.
- Built **CloudComputeManager** for GPU spot-instance provisioning, environment setup, pipeline execution, preemption recovery, and per-run cost tracking; used it for MXene shear sweeps.

Manufacturing & Production Systems

Engineering Contractor, Rocky Mountain Scientific Laboratory

Jun–Aug 2026

Quality system for a client affiliate's new energetic-materials production line

Automation & Robotics embed, Jun–Aug 2026

- Developed lot-release records covering inspections, approval authority, hold criteria, and nonconformance disposition, with item-level traceability to the governing **military specification**.
- Built a controlled, single-source document set with **10 client-format instructions**, **20 test-specific quality-control packets**, manufacturing procedures, and inspections.
- Automated checks for process-route isolation, link validity, terminology, and consistency between released documents and their source.
- Built a program tracker that generates the Kanban board, decision log, requirements traceability matrix, and design-review record.
- Supported quality, packaging, labeling, on-site commissioning, and hazard and failure-mode analysis.

National Security & Technology Transition

S&T Scouting Intern

The Joint Staff, J7 / Future Technology Office, Aug–Oct 2025

- Produced **eight technical briefs** on propulsion and energy concepts, assessing readiness, risk, and experimentation paths for TRL 3–5 technologies across Service laboratories and DoD R&D organizations.

Technology Commercialization Intern

Idaho National Laboratory (DOE), May–Aug 2024

- Conducted **20 Energy I-Corps interviews**, developed a commercialization plan for a laboratory technology, and supported a DOE proposal awarded **\$2M**.

Education

University of Colorado Boulder

Expected May 2028

Ph.D., Materials Science & Engineering. Advisor: Prof. Hendrik Heinz.

University of Florida

May 2024

B.S., Materials Science & Engineering. Minor in Chemistry.

Core Skills

Materials & Modeling: LAMMPS; NAMD; VASP/VASPsol; Quantum ESPRESSO; ASE; pymatgen; MDAnalysis; force-field parameterization; elastic and surface-energy validation.

Research Automation: Campaign orchestration; resource scheduling; adapters for a simulated lab, local compute, and human tasks; run provenance; restart recovery; experiment-selection workflows; policy checks.

Software & HPC: Python; Bash; Linux; Git; Docker; SQLite; pytest; CI; Slurm; PBS Pro; checkpoint/restart; GPU and CPU-MPI workflows.

ML & Data: scikit-learn; TensorFlow; RDKit; SOAP descriptors; regression and classification; surrogate modeling; uncertainty analysis.

Manufacturing & Quality: Lot-release records; military-specification traceability; controlled procedures; nonconformance disposition; hazard and failure-mode analysis; bills of materials; vendor qualification.

Selected Research Outputs

Co-author: *INTERFACE Force Field for Alumina with Validated Bulk Phases and a pH-Resolved Surface Model Database for Electrolyte and Organic Interfaces*. [arXiv:2601.12570](https://arxiv.org/abs/2601.12570), 2026 (preprint).

Lead-author manuscripts in preparation: *Modeling of MXene Shear Properties via Atomistic Molecular Dynamics Simulations*, with AFRL Materials & Manufacturing Directorate co-authors;

Understanding Hydrogen Release Mechanisms of N-Ethylcarbazole on Pt Nano-Crystal Catalysts.